Practice workbook · Simulations and models: why we use them and their limits

Why scientists and others use computer simulations, their benefits, problems with relying on them, and their processing demands.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-14

1. Question 4 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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1. Question 4 · part a · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

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(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

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(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

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2. Question 4 · part a · 1 mark · Working and answer

Working and answer

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3. Question 4 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

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.....

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(b) Explain **one** reason why a simulation may require a powerful processor.

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(c) A scientist uses Boolean logic in programs.

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1	1	1			

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3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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4. Question 3 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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4. Question 3 · part a · 2 marks · Working and answer

Working and answer

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